

MALINDI, Kenya

WMO: 637990

Lat: 3.229S

Lon: 40.102E

Elev: 82

StdP: 14.65

Time Zone: 3.00 (E03)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	71.6	73.1	64.6	91.5	78.1	66.2	96.9	77.8	24.4	80.6	22.9	80.8	4.1	180	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	10.8	90.0	78.8	89.7	78.6	89.2	78.5	80.8	87.1	80.1	86.4	79.8	86.1	14.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	151.3	84.9	78.7	149.3	84.6	77.5	143.6	83.9	44.5	87.5	43.9	87.0	43.5	86.7	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	21.0	19.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.1	82.0	82.7	83.9	83.9	80.5	79.4	78.3	77.8	79.0	80.4	82.0	82.9
	DBStd	2.69	1.41	1.34	1.53	2.02	2.14	1.66	1.34	2.67	1.93	1.50	1.60	1.32
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11335	992	915	1050	1017	945	883	876	863	869	943	961	1021
	CDD65	5861	527	495	585	567	480	433	411	399	419	478	511	556
	CDH74	59523	5662	5544	7229	6954	4584	3774	2926	2912	3320	4478	5613	6527
	CDH80	17103	1681	1832	2958	2799	998	540	247	289	510	1071	1802	2376
Wind	WSAvg	10.4	11.6	11.0	8.9	10.5	10.7	12.1	12.2	11.6	11.2	9.5	6.8	8.4
Precipitation	PrecAvg	41.0	0.4	0.1	1.7	6.8	12.0	6.5	4.3	2.6	1.8	3.7	2.8	1.3
	PrecMax	67.3	3.6	1.2	10.8	16.9	23.7	37.5	16.1	9.4	6.8	24.1	9.3	4.3
	PrecMin	26.1	0.0	0.0	0.0	0.2	2.7	0.6	0.4	0.4	0.0	0.0	0.0	0.0
	PrecStd	10.1	0.7	0.2	2.2	4.0	4.9	5.6	2.9	1.9	1.6	5.0	2.3	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.5	90.0	91.7	91.6	88.1	85.5	84.1	84.3	85.8	88.0	89.9	90.0
		MCWB	78.3	77.8	79.4	79.4	78.6	76.4	74.7	74.1	75.5	76.9	78.0	79.2
	2%	DB	88.1	89.1	90.0	90.2	86.3	84.4	82.7	82.8	84.4	86.3	88.2	89.5
		MCWB	77.6	77.6	78.7	79.0	77.8	76.1	74.2	73.8	74.8	76.4	78.4	78.9
	5%	DB	87.6	88.0	89.7	89.7	85.7	83.9	82.3	82.5	83.9	85.9	87.8	88.1
		MCWB	77.4	77.2	78.6	78.9	77.5	75.8	74.0	73.8	74.5	76.3	78.6	78.6
	10%	DB	86.3	87.5	88.9	88.7	84.4	82.7	81.0	82.0	82.6	84.5	86.4	87.7
		MCWB	76.7	77.0	78.5	78.9	76.9	75.0	73.4	73.7	74.0	75.8	78.2	78.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	79.7	81.2	81.4	80.1	78.8	76.5	76.5	77.5	79.2	80.1	81.7
		MCDB	86.1	87.2	88.3	87.4	85.4	83.4	81.6	81.1	82.6	84.4	86.5	87.3
	2%	WB	79.7	78.8	80.3	80.5	79.1	77.2	75.5	75.6	76.4	78.3	79.8	80.6
		MCDB	85.3	86.3	87.1	87.0	84.7	82.5	81.0	80.5	81.8	83.7	86.1	86.7
	5%	WB	78.6	78.1	79.9	80.0	78.2	76.4	74.8	74.7	75.6	77.7	79.3	79.9
		MCDB	84.5	85.5	86.6	86.5	83.7	81.8	80.4	80.0	81.1	82.9	85.2	86.0
	10%	WB	77.6	77.4	79.4	79.6	77.6	75.9	74.3	74.3	75.0	77.0	78.9	79.2
		MCDB	84.1	84.7	86.0	86.1	82.8	81.4	79.9	79.7	80.6	82.2	84.5	85.2
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.1	10.8	9.0	7.9	7.2	7.1	8.6	8.6	9.5	9.8	10.3
		MCDBR	10.8	10.8	11.7	10.0	8.7	7.8	7.5	8.6	8.9	10.4	10.4	10.8
		MCWBR	4.1	3.6	4.1	3.4	4.0	4.3	4.2	4.3	3.8	3.8	4.0	4.4
	5% WB	MCDBR	10.5	10.3	10.5	9.2	8.3	7.5	7.4	8.5	8.7	9.3	9.9	10.7
		MCWBR	4.5	3.9	4.2	3.6	4.3	4.4	4.4	4.9	4.5	4.0	4.2	4.7
Clear-Sky Solar Irradiance	taub		0.505	0.512	0.468	0.418	0.435	0.464	0.456	0.433	0.419	0.431	0.431	0.481
	taud		2.148	2.124	2.295	2.486	2.415	2.275	2.290	2.370	2.408	2.394	2.415	2.247
	Ebn at Noon		266	265	274	280	267	254	259	272	283	284	285	271
	Edh at Noon		51	53	44	35	36	41	41	39	39	40	39	46
All-Sky Solar Radiation	RadAvg		1915	1995	2003	1812	1563	1541	1583	1736	1911	1961	1953	1931
	RadStd		70	59	86	107	118	63	74	66	74	105	88	75

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.65	+0.85	+0.83	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page